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An Optimal Parking Path Planning Method Integrating Motion Mode Decision-making for 4WIS Vehicles*

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***Abstract*— Four-wheel independent steering (4WIS) vehicles demonstrate high flexibility and maneuverability, rendering them as promising platforms for autonomous driving technologies. However, existing motion planning methods for 4WIS vehicles ignored their multiple motion modes, thus failing to fully leverage their excellent maneuver advantages in narrow scenarios. To address this issue, this paper proposes an optimal parking path planning integrating motion mode decision-making approach. Firstly, a coarse path planning is conducted based on the modified A* algorithm, facilitating the formulation and solution of the optimization problem. Then, a path planning-oriented simplified vehicle model is designed to reduce problem complexity and alleviate computational burden. Next, a switched affine system for multiple motion modes is constructed, and the logics associated with motion modes are introduced to formulate the parking path planning problem as a mixed-integer programming (MIP) task. Finally, the MIP is solved to obtain the optimal path and motion mode decisions simultaneously. Simulation results demonstrate that the proposed algorithm effectively leverages the flexibility and maneuverability advantages of 4WIS vehicles in parking scenarios.**

***Index Terms*— 4WIS vehicle, Autonomous parking, Path planning, Multiple motion modes**

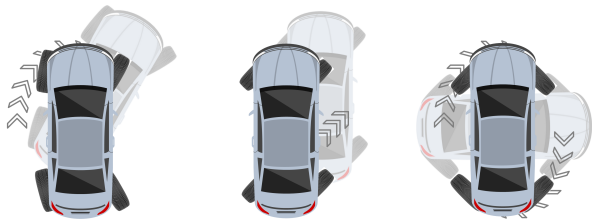
I. INTRODUCTION

In recent years, autonomous driving technology, coupled with advancements in electric vehicles [1], has rapidly emerged as a focal point of research in the automotive field [2]. Vehicles configured with four-wheel independent steering (4WIS) hold promise as a novel platform for the autonomous driving technology, owing to their excellent maneuverability and flexibility [3]. The 4WIS vehicles possess multiple motion modes, which can be categorized into the following three types (see Fig. 1): 1) *Ackermann steering mode*, the wheels of both the front and rear axles satisfying the Ackermann steering angle relationship; 2) *Diagonal motion mode*, all wheels oriented in the same direction, allowing the vehicle to translate along a diagonal line; 3) *Zero turn mode*, the vehicle rotating in place with a 0 steering radius. By combining these different motion modes, vehicles can maneuver flexibly in narrow scenarios, such as parking situations with obstacles.

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(a) (b) (c)

Fig. 1: Multiple motion modes for 4WIS vehicles. (a) Ackermann steering mode; (b) Diagonal motion mode; (c) Zero turn mode.

Motion planning is a critical component within the framework of autonomous driving technology, usually regarded as the coupling of path planning and speed planning. Its primary function is to assist vehicles in finding a collision-free, smooth, and efficient desired trajectory, incorporating a series of spatio-temporal information. This study primarily focuses on the autonomous parking path planning of 4WIS vehicles, where their flexibility and maneuverability can be fully demonstrated.

A. Related Works

1) Autonomous parking motion planning: Fruitful studies have been reported on autonomous parking motion planning, which can be mainly categorized into three types: geometric-based, search-and-sampling-based, and optimization-based methods. The geometric-based approach primarily utilizes specific curves, such as Dubins [4] and Reeds-Shepp [5], to establish connections between the vehicle and parking spaces. The search-and-sampling-based approaches primarily involve generating sample points in state space or control space and subsequently selecting a path from these samples [6]. These methods mainly encompass rapidly-exploring random trees [7], probabilistic roadmaps [8], and hybrid A* [9]. On the other hand, optimization-based methods, as exemplified in [6], typically formulate motion planning as an optimal control problem, thereby obtaining collision-free optimal trajectories and action sequences. Although parking planning for front-wheel steering vehicles has become increasingly mature, there are still various aspects to be explored in the motion planning of 4WIS vehicles.

2) Motion planning for 4WIS vehicles: Researchers have extensively investigated the motion planning issues pertaining to 4WIS vehicles. Reference [10] addressed the problem of active obstacle avoidance path planning with the clothoid

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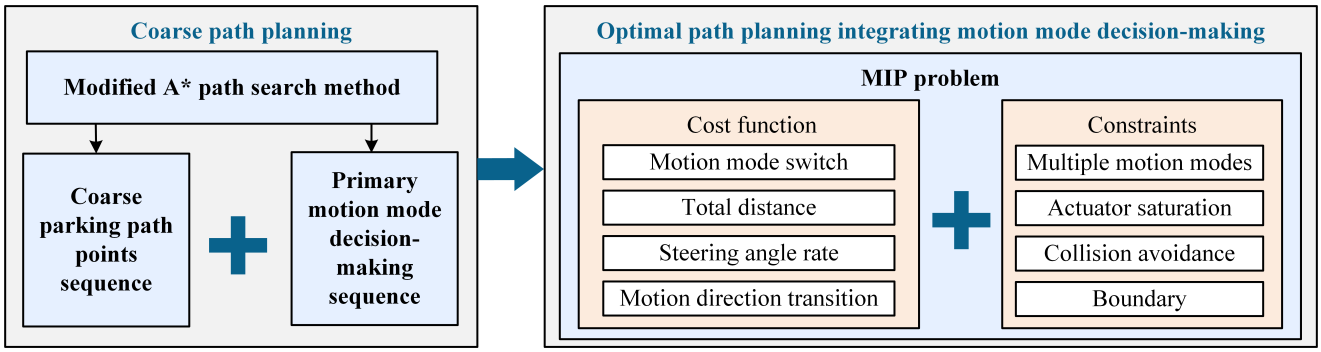


Fig. 2: Overall framework of the proposed parking path planning method for 4WIS vehicles.

method. Howevere, the multiple motion modes characteristic of 4WIS was ignored. In addressing the intermediate dynamic process of steering modes switching of 4WIS vehicle, [11] proposed a trajectory planning method based on B-spline curves, but did not investigate the decision-making of steering modes. For the hign-speed 4WIS robots, [12] employs the A* algorithm for path planning. Similarly, [13] proposed the shape-aware lifelong planning A* algorithm for addressing the maneuver issue of differential steering vehicles capable in narrow scenarios.

3) Mixed Logical Dynamical Systems: To characterize the motion dynamics of 4WIS vehicle, it can be regarded as a mixed logical dynamical (MLD) system, comprising discrete and continuous control inputs [14]. Previous studies have presented various applications of MLD systems. In [15], an hybrid model predictive controller (HMPC) based on MLD system was proposed for vehicle cruise control, where the gears were considered as binary variables. In [16], an optimal longitudinal control method considering vehicle drive and brake mode switching was proposed, treating the nonlinear vehicle longitudinal dynamics system as an MLD system. In [17], to address the task of planar manipulation of robot, a convex HMPC was designed to obtain the optimal multiple contact modes associated with frictional interactions. However, research on trajectory or path planning for MLD systems is limited.

B. Motivations and Contributions

The aforementioned studies on motion planning for 4WIS vehicles have achieved satisfactory results. However, when it comes to addressing parking planning issues for 4WIS vehicles, conventional methods exhibit certain limitations, particularly in narrow parking scenarios. On one hand, the multiple motion modes of 4WIS vehicles are usually ignored in parking motion planning, much less the decision-making of motion modes. On the other hand, the characteristics of 4WIS vehicles as MLD systems are not fully utilized, lacking a unified modeling approach to represent its motion dynamic. Therefore, the high maneuverability and flexibility characteristics of 4WIS vehicles have not been fully exploited. To the best of our knowledge, scant research has been conducted on the parking motion planning issues concerning 4WIS vehicles with multiple motion modes.

In response to these issues, we propose an optimal parking path planning method integrating motion mode decision-making for 4WIS vehicles. The schematic of the overall algorithm framework is depicted in Fig. 2. We propose a coarse path search method to prepare for subsequent optimization-based planning. To reduce the complexity of problem, we decouple the trajectory's spatio-temporal information and introduce the path planning-oriented simplified vehicle models. By fully utilizing the characteristics of MLD systems to handle various logical propositions, we formulate the parking path planning problem as a mixed-integer programming (MIP) problem. The solution of the MIP corresponds to the optimal path and the sequence of optimal motion modes desion.

The principal contributions of this paper are as follows:

- A coarse path search method based on modified A* algorithm is proposed to facilitate specifying the constraints and provide initial guess for MIP. The rules for node expansion are redefined for 4WIS vehicles, and the cost function is modified to quickly find feasible coarse paths. Compared with the other methods [10]–[12], the muiltple motions modes are fully exploited.
- The path planning-oriented simplified vehicle models are established to reduce the complexity of the planning problem and alleviate computational burden. By introducing approximate distance variables, the spatio-temporal information is decoupled, and the intermediate dynamics between different motion modes are not required to be considered. Compared with trajectory planning method [18], it facilitates extending optimization-based method to address the motion planning problem for the complex dynamic of 4WIS vehicles.
- The parking path planning problem for 4WIS vehicles is formulated as an MIP, considering continuous/discrete control inputs and the logics associated with multiple motion modes. Compared to other methods [10]–[13], [18], it fully considers the vehicle kinematic characteristics under multiple motion modes and ensures the optimality of both the paths and motion modes.

C. Organization of the paper

The remainder of this paper is organized as follows: Section II designs the coarse path planning method. The path

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planning-oriented simplified vehicle models and the MIP problem formulation are proposed in Section III. Simulation results and discussion are presented in Section IV. Finally, Section V concludes this paper.

II. Coarse Path Planning

Our study leverages the A* algorithm to rapidly generate coarse paths. Several modifications are made to the traditional A* algorithm to accommodate the parking maneuver of 4WIS vehicles.

First, to accurately characterize the vehicular maneuvers in parking scenarios, the yaw angle of the vehicle is incorporated into the node state. This adjustment facilitates more precise collision check, consequently improving the quality of A* path planning results.

Second, the node expansion rules have been redefined, and logic for eliminating low-quality child nodes has been incorporated. In Fig. 3, the expansion rules for child nodes in two representative scenarios are illustrated. These rules account for all motion modes, fully manifesting the high flexibility of 4WIS vehicles. Moreover, various selection criteria have been introduced during node expansion. For instance, to ensure continuity and smoothness of maneuvers, the distance between adjacent waypoints in zero turn mode is maintained at an appropriate level. Additionally, under Ackermann steering mode, frequent changes in heading are restricted to prevent excessively large curvature.

Third, the cost function of nodes is modified as follows:

$$\begin{aligned} f &= g + 2h \\ h &= |x_{curr} - x_{end}| + |y_{curr} - y_{end}| + |\varphi_{curr} - \varphi_{end}| \end{aligned}, \quad (1)$$

where g represents the cumulative cost from the start node to the current node, h represents the estimated cost from the current node to the goal node, $[x_{curr}, y_{curr}, \varphi_{curr}]$ denotes the current position and heading of vehicle, and $[x_{end}, y_{end}, \varphi_{end}]$ denotes the target position and heading. As the cost value g is continuously accumulated, we only focus on the cost value from the parent node to the current node, denoted as $g_{parent-curr}$. Its calculation adheres to the following rules:

$$g_{parent-curr} = \begin{cases} g_1 & \text{if zero turn or straight move} \\ g_2 & \text{else if diagonal motion} \\ g_3 & \text{else if Akermann steering} \end{cases}, \quad (2)$$

with g_i (where $i \in 1,2,3$) specified according to the current motion mode. The purpose of these definitions is to rapidly search for feasible coarse paths, albeit at the expense of optimality. Fortunately, subsequent optimal path planning integrating motion mode decision-making will be conducted based on MIP, rendering optimality unnecessary during the coarse path planning stage.

Fourth, the termination condition for the search has been modified. It is now considered that the search process concludes when the vehicle's body contour enters the parking space without collision in some narrow scenarios, rather than requiring arrival at the target node. This adjustment aims to reduce the time spent on path searching.

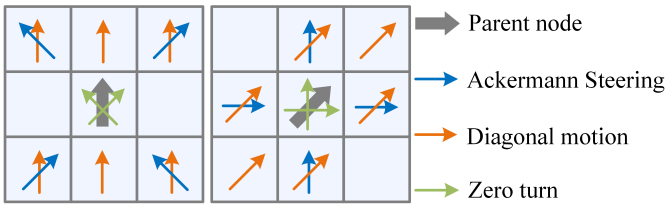


Fig. 3: Node Expansion Methods.

Finally, we will preliminarily fine-tune the search results in preparation for subsequent optimization-based path planning. If an Ackermann steering node is sandwiched between other modes, it can be considered as an zero turn point. Subsequently, the path is divided into several segments based on the zero turn points. Equal-distance interpolation is applied to each segment (including heading angle changes during zero turn point), such that the total number of interpolated path points is $N+1$. The interpolated coarse path points are denoted as $[x_{coarse}(k), y_{coarse}(k), \varphi_{coarse}(k)], k \in \mathbb{I}_0^{N-1}$.

III. Optimal Path Planning Integrating Motion Mode Decision-Making

A. Path Planning-Oriented Simplified Vehicle Models

In the process of parking, due to the relatively low vehicle speed, kinematic models have demonstrated high precision in describing vehicle dynamics. Therefore, this paper opts for kinematic models to formulate the three types of motion modes:

$$\begin{cases} \dot{x}(t) = v_r(t)\cos\delta_r(t)\cos\varphi(t) \\ \dot{y}(t) = v_r(t)\cos\delta_r(t)\sin\varphi(t) \\ \dot{\varphi}(t) = -v_r(t)/L_r \cdot \sin\delta_r(t) \\ \dot{v}_r(t) = a_r(t)\cdot \end{cases}, \quad (3)$$

$$\begin{cases} \dot{x}(t) = v_r(t)\cos(\varphi(t)+\delta_r(t)) \\ \dot{y}(t) = v_r(t)\sin(\varphi(t)+\delta_r(t)) \\ \dot{\varphi}(t) = 0 \\ \dot{v}_r(t) = a_r(t)\cdot \end{cases}, \quad (4)$$

$$\begin{cases} \dot{x}(t) = 0 \\ \dot{y}(t) = 0 \\ \dot{\varphi}(t) = 2v_r(t)/W \cdot \cos\left(\arctan\left(\frac{2L_r}{W}\right)\right) \\ \dot{v}_r(t) = a_r(t)\cdot \end{cases}, \quad (5)$$

where $t \in [0,T]$ represents the parking maneuver duration, (x,y) denotes the vehicle's center of gravity (CoG), φ is yaw angle, v_r and a_r are the linear velocity and acceleration of the rear wheel, δ_r is the steering angle of the rear wheel in single-track model, L_r denotes the distance from CoG to the rear axis, and W is vehicle's width. For the Ackermann steering mode, this study requires that the line connecting the steering center with the vehicle's CoG must be perpendicular to the vehicle's longitudinal axis. Regarding the zero turn mode, only the scenario where the vehicle rotates around its CoG is considered for simplification.

In this paper, a path planning-oriented simplification of the original kinematic model is proposed, considering the following two reasons: 1) Trajectory planning for the spatio-temporal dynamics of three motion modes is challenging.

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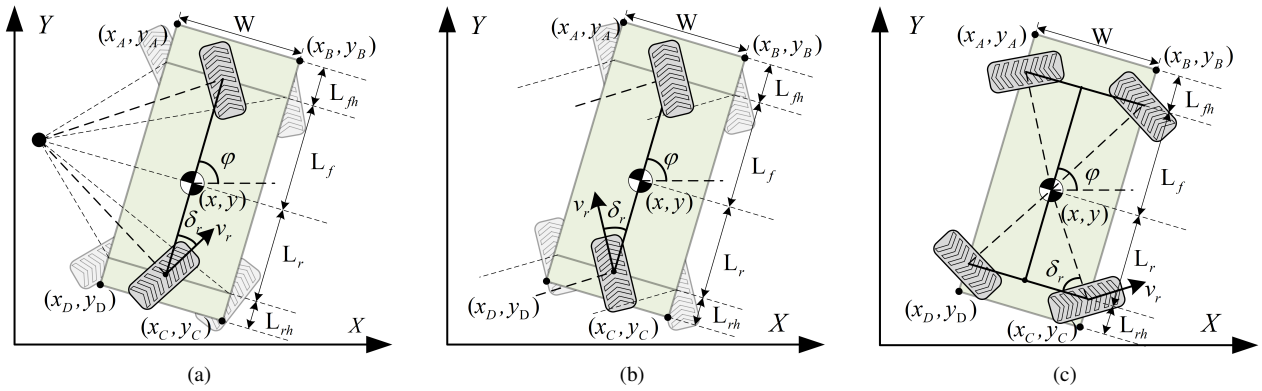


Fig. 4: Schematics of vehicle kinematic models for multiple motion modes. (a) Ackermann steering mode; (b) Diagonal motion mode; (c) Zero turn mode.

Addressing path planning separately implies fewer variables and constraints; 2) For safety reasons, it is stipulated to halt the vehicle before motion mode switching. However, representing the intermediate dynamics between two modes in the optimization problem is difficult. Thus, decoupling spatio-temporal dynamics for path planning can be adopted. Subsequently, more intricate considerations such as detailed speed planning and smooth mode switching control may be explored after the path planning. However, these aspects are beyond the scope of this study. This paper primarily focuses on the path planning probelm. The simplifications oriented towards path planning for (3)-(5) are as follows:

$$\begin{cases} x(k+1) = x(k) + p(k)\cdot\Delta d\cos\delta_r(\mathrm{k})\cos\varphi(k) \\ y(k+1) = y(k) + p(k)\cdot\Delta d\sin\delta_r(\mathrm{k})\sin\varphi(k) \\ \varphi(k+1) = \varphi(k) - p(k)\cdot\Delta d/L_r\cdot\sin\delta_r(\mathrm{k}) \end{cases}, \tag{6}$$

$$\begin{cases} x(k+1) = x(k) + p(k)\cdot\Delta d\cos\left(\varphi(k)+\delta_r(\mathrm{k})\right) \\ y(k+1) = y(k) + p(k)\cdot\Delta d\sin\left(\varphi(k)+\delta_r(\mathrm{k})\right) \\ \varphi(k+1) = \varphi(k) \end{cases}, \tag{7}$$

$$\begin{cases} x(k+1) = x(k) \\ y(k+1) = y(k) \\ \varphi(k+1) = \varphi(k) - 2p(k)\cdot\frac{\Delta d}{W}\cos\left(\arctan\left(\frac{2L_r}{W}\right)\right) \end{cases}, \tag{8}$$

where $k \in \mathbb{I}_0^{N-1}$ is the index of path points, $p(k) \in \{-1, 1\}$ denotes the direction of motion, and Δd is the virtual interval between adjacent path points. In the Ackermann steering mode and the diagonal motion mode, $p = -1$ and $p = 1$ represent reverse and forward, respectively; in the zero turn mode, $p = -1$ and $p = 1$ represent counterclockwise and clockwise rotation, respectively. The principles of simplification are detailed below.

In many trajectory planning studies, the time horizon T is uniformly discretized into $N+1$ sampling instants, denoted as $\{0, \Delta T, 2\Delta T, \ldots, N\Delta T\}$. Due to the coupling of temporal information in the kinematic model, direct path planning is not feasible. In this paper, an approach utilizing non-uniform time intervals, denoted as $\{0, \Delta T'(0), \sum_{i=0}^{1}\Delta T'(i), \ldots, \sum_{i=0}^{N-1}\Delta T'(i)\}$, is adopted as opposed to the other method. Note that $T = \sum_{i=0}^{N-1}\Delta T'(i)$. We establish the equality

$$v_r(k)\cdot\Delta T'(k) \equiv \Delta d = d/N,\ k \in \mathbb{I}_0^{N-1}, \tag{9}$$

where d represents the approximate total distance of path planning. This implies that we do not concern about how $v_r(k)$ changes, as there always exists a $\Delta T'(k)$ to adapt and generate a Δd. The velocity planning and path planning are fully decoupled, and velocity planning can be performed based on the results of path planning. Simplification and discretization of (3)-(5) by applying (9), we can get path planning-oriented simplification (6)-(8). For ease of expression in subsequent discussions, they are abbreviated as follows:

$$\zeta(k+1) = f_i(\zeta(k), u(k)) \ , \tag{10}$$

where $\zeta(k) = [x(k), y(k), \varphi(k)]$, $u(k) = [p(k), \Delta d, \delta_r(k)]$, and $i \in \{1, 2, 3\}$ denotes indices for the three different motion modes.

B. Constraints Construction

1) Multiple motion modes constraints: The following biconditional propositions are defined to characterize the motion states:

$$\begin{array}{l} \alpha_1(k) = 1 \leftrightarrow \text{Akermann steering mode at } k_{\text{th}} \text{ point} \\ \alpha_2(k) = 1 \leftrightarrow \text{Diagonal motion mode at } k_{\text{th}} \text{ point} \\ \alpha_3(k) = 1 \leftrightarrow \text{Zero turn mode at } k_{\text{th}} \text{ point} \end{array}, \tag{11}$$

where $\alpha_1(k)$, $\alpha_2(k)$, and $\alpha_3(k) \in \{0, 1\}$ are binary variables. At any given moment within time horizon T, the vehicle can travel adhering to only one motion mode, i.e.,

$$\alpha_1(k) + \alpha_2(k) + \alpha_3(k) = 1, k \in \mathbb{I}_0^{N-1}. \tag{12}$$

The vehicle kinematic models of multiple motion modes can be represented by switched affine system (SAS), which is rewritten as

$$\zeta(k+1) = \zeta_1(k+1) + \zeta_2(k+1) + \zeta_3(k+1), k \in \mathbb{I}_0^{N-1} \tag{13}$$

with the if-then-else conditions

$$\zeta_i(k+1) = \begin{cases} f_i(\zeta(k), u(k)), & \text{if } \alpha_i(k) = 1 \\ 0, & \text{otherwise} \end{cases}, \tag{14}$$

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where $\zeta_i(k+1) = [x_i(k+1), y_i(k+1), \varphi_i(k+1)]$ represents the intermediate virtual state corresponding to the $(k+1)_{\text{th}}$ path point derived according to (14).

To incorporate constraints into the optimization problem, it is necessary to transform logical expressions into mixed-integer inequalities. In this paper, the Big M method is adopted to convert the if-then-else conditions in (14) to inequalities, i.e.,

$$\begin{cases} \zeta_i(k+1) \geq m_\zeta(k+1)\alpha_i(k) \\ \zeta_i(k+1) \leq M_\zeta(k+1)\alpha_i(k) \\ \zeta_i(k+1) - f_i(\zeta(k), u(k)) \geq -M_\zeta(k+1)(1-\alpha_i(k)) \\ \zeta_i(k+1) - f_i(\zeta(k), u(k)) \leq -m_\zeta(k+1)(1-\alpha_i(k)) \end{cases}, \tag{15}$$

where $i \in \{1,2,3\}$, $m_\zeta(k+1) = [m_x(k+1), m_y(k+1), m_\varphi]$ is the lower bound of $\zeta_i(k+1)$, and $M_\zeta(k+1) = [M_x(k+1), M_y(k+1), M_\varphi]$ is the upper bound of $\zeta_i(k+1)$. The boundaries m_x, M_x, m_y, and M_y are computed from coarse path points:

$$\begin{aligned} &m_x(k) = x_{coarse}(k) - \beta_x, M_x(k) = x_{coarse}(k) + \beta_x, \\ &m_y(k) = y_{coarse}(k) - \beta_y, M_y(k) = y_{coarse}(k) + \beta_y, \\ &m_\varphi(k) = \varphi_{coarse}(k) - \beta_\varphi, M_\varphi(k) = \varphi_{coarse}(k) + \beta_\varphi, \end{aligned} \tag{16}$$

where β_x, β_y, and β_φ are constant that reflect the degree of trust in the coarse path.

2) Actuator saturation constraints: For ensuring that the vehicle can perform zero turns, the maximum achievable steering angle for the left rear/right rear wheel should satisfy

$$|\overline{\delta_{rl}}| = |\underline{\delta_{rl}}| = |\overline{\delta_{rr}}| = |\underline{\delta_{rr}}| \geq \arctan(\frac{2L_r}{W}), \tag{17}$$

where $\overline{(\cdot)}$ and $\underline{(\cdot)}$ represent the upper and lower bounds, respectively, and the subscripts "rl" and "rr" denote the left rear wheel and right rear wheel, correspondingly.

As for the simplified single-track models of Ackermann steering and diagonal motion modes (see Fig. 4(a) and 4(b)), there are slight differences in the range of virtual rear wheel steering angle δ_r. For the diagonal motion mode, the geometric relationship is as follows:

$$\delta_{rl}(k) = \delta_{rr}(k) = \delta_r(k). \tag{18}$$

However, from geometric relations of Ackermann steering mode, it is derived that

$$\begin{aligned} \delta_{rl}(k) &= \arctan\left(\frac{2L_r \tan\delta_r(k)}{2L_r + W\tan\delta_r(k)}\right) \\ \delta_{rr}(k) &= \arctan\left(\frac{2L_r \tan\delta_r(k)}{2L_r - W\tan\delta_r(k)}\right) \end{aligned}. \tag{19}$$

Hence, when the steering angle reaches its extremum under the Ackermann steering mode, it can be deduced that

$$|\overline{\delta_r}| = |\underline{\delta_r}| = \arctan(\frac{L_r}{W}). \tag{20}$$

Consequently, the following logical proposition holds:

$$\begin{cases} -\arctan(\frac{L_r}{W}) \leq \delta_r(k) \leq \arctan(\frac{L_r}{W}), & \text{if } \alpha_1(k) = 1 \\ -\arctan(\frac{2L_r}{W}) \leq \delta_r(k) \leq \arctan(\frac{2L_r}{W}), & \text{otherwise} \end{cases}. \tag{21}$$

Convert the proposition to mixed-integer inequalities, i.e.,

$$\begin{cases} \delta_r(k) \geq (\arctan(\frac{2L_r}{W}) - \arctan(\frac{L_r}{W})) \cdot \alpha_1(k) - \arctan(\frac{2L_r}{W}) \\ \delta_r(k) \leq \arctan(\frac{2L_r}{W}) - (\arctan(\frac{2L_r}{W}) - \arctan(\frac{L_r}{W})) \cdot \alpha_1(k) \end{cases}. \tag{22}$$

3) Collision avoidance constraints: In this study, the safe travel corridors method (see [19]) is leveraged to generate collision avoidance constraints. Two discs covring the vehicle body are adopted to design the constrints, wherein their coordinates of centers are denoted as $(x_{d1}(k), y_{d1}(k))$ and $(x_{d2}(k), y_{d2}(k))$. Starting from centers of discs, two series of corridors along the coarse path are gradually generated by expanding from small squares in four directions, denoted as $\{STC_i(0), STC_i(1), \ldots, STC_i(N)\}$, $i \in \{1,2\}$. Constrain the vehicle within such corridors, i.e.,

$$\begin{cases} (x_{d1}(k), y_{d1}(k)) \in STC_1(k) \\ (x_{d2}(k), y_{d2}(k)) \in STC_2(k) \end{cases}. \tag{23}$$

Each variable has a certain degree of freedom in variation, allowing the optimizer to continue seeking optimal paths and mode decisions within these corridors.

4) Boundary constraints: Constrain the initial and final positions of parking:

$$\begin{cases} [x(0), y(0), \varphi(0)] = [x_{start}, y_{start}, \varphi_{start}] \\ rect_{ABCD}(N) \subset \Gamma_{parking} \\ \varphi_{end} - \gamma < \varphi(N) < \varphi_{end} + \gamma \\ \delta_r(N-1) = 0 \end{cases}, \tag{24}$$

where $[x_{start}, y_{start}, \varphi_{start}]$ denotes the initial pose of the vehicle, $rect_{ABCD}(N)$ represents the shape contour of vehicle body, $\Gamma_{parking}$ is the target parking area, and $\gamma > 0$ is a constant. The coordinates of the four vertices of $rect_{ABCD}(N)$ are directly derived from the CoG and $\varphi(N)$, where further details are omitted for brevity. Compared to restricting the final position of the vehicle to a single point, constraining it within a range is slacker and easier to solve.

C. Cost Function Design

1) Motion mode switch cost: The number of mode switches, denoted as N_{switch}, among $N+1$ path points can be calculated:

$$N_{switch} = \sum_{i=1}^{3} \sum_{j=0}^{N-2} \frac{(\alpha_i(j+1) - \alpha_i(j))^2}{2}. \tag{25}$$

Therefore, the mode switching cost J_1 is defined aims to minimize the frequency of mode changes as much as possible:

$$J_1 = \lambda_1 \cdot N_{switch}, \tag{26}$$

where $\lambda_1 > 0$ is the weighting factor of J_1.

2) Total distance cost: In Section III-A, since the spatio-temporal decoupling method removes temporal information, the path length is chosen to be penalized. Note that the virtual total distance d is merely an approximation. Thus, we use the cumulative distance between path points to represent the total path length:

$$J_2 = \lambda_2 \cdot \sum_{i=0}^{N-1} (x(i+1) - x(i))^2 + (y(i+1) - y(i))^2, \tag{27}$$

where $\lambda_2 > 0$ is the weighting factor of J_2.

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3) Steering angle rate cost: The rate of steering angle needs to be penalized for smoothness. However, when it comes to mode switching, the steering angle δ_r between different modes may suddenly jump. For example, in the diagonal motion mode, suppose the vehicle travels with the maximum steering angle, i.e. $\delta_r = \arctan(\frac{2L_r}{W})$. When switching to the Ackermann steering mode, since the maximum steering angle becomes $\arctan(\frac{L_r}{W})$, the change in δ_r before and after the switching point must inevitably undergo a change of at least $\Delta\delta_r = \arctan(\frac{2L_r}{W}) - \arctan(\frac{L_r}{W})$. To avoid this situation, auxiliary variables $\delta_{r1}(k)$ and $\delta_{r2}(k)$, $k \in \mathbb{I}_0^{N-1}$ are designed, following the logic outlined below:

$$\delta_{ri}(k) = \begin{cases} \delta_r(k), & \text{if } \alpha_i(k) = 1 \\ 0, & \text{otherwise} \end{cases}, i \in \{1,2\}. \tag{28}$$

Convert (28) to mixed-integer inequalities by Big M method:

$$\begin{cases} \delta_{r1}(k) \geq -\arctan(\frac{L_r}{W}) \cdot \alpha_1(k) \\ \delta_{r1}(k) \leq \arctan(\frac{L_r}{W}) \cdot \alpha_1(k) \\ \delta_{r1}(k) - \delta_r(k) \geq -\arctan(\frac{L_r}{W}) \cdot (1-\alpha_1(k)) \\ \delta_{r1}(k) - \delta_r(k) \leq \arctan(\frac{L_r}{W}) \cdot (1-\alpha_1(k)) \end{cases}, \\ \begin{cases} \delta_{r2}(k) \geq -\arctan(\frac{2L_r}{W}) \cdot \alpha_2(k) \\ \delta_{r2}(k) \leq \arctan(\frac{2L_r}{W}) \cdot \alpha_2(k) \\ \delta_{r2}(k) - \delta_r(k) \geq -\arctan(\frac{2L_r}{W}) \cdot (1-\alpha_2(k)) \\ \delta_{r2}(k) - \delta_r(k) \leq \arctan(\frac{2L_r}{W}) \cdot (1-\alpha_2(k)) \end{cases}. \tag{29}$$

The steering angle rate cost is designed as

$$J_3 = \lambda_3 \cdot \sum_{i=1}^{2} \sum_{j=0}^{N-2} (\delta_{ri}(j+1) - \delta_{ri}(j))^2, \tag{30}$$

where $\lambda_3 > 0$ is the weighting factor of J_3.

4) Motion direction transition cost: The motion direction transition cost is defined as following:

$$J_4 = \lambda_4 \cdot \sum_{i=0}^{N-2} (p(i+1) - p(i))^2, \tag{31}$$

where $\lambda_4 > 0$ is the weighting factor of J_4.

D. Mixed-integer Programming Problem

The optimal path planning integrating motion mode decision-making is constructed as the following mixed-integer programming problem:

$$\begin{aligned} &\arg\min_{\boldsymbol{I},\boldsymbol{C}} J = \sum_{i=1}^{4} J_i \\ &s.t. \quad \begin{cases} \text{Multiple motion modes constraints } (12),(13),(15); \\ \text{Actuator saturation constraints } (22); \\ \text{Collision avoidance constraints } (23); \\ \text{Boundary constraints } (24); \\ \text{Other constraints } (29). \end{cases} \end{aligned} \tag{32}$$

All integer and continuous variables are denoted as $\boldsymbol{I}$ and $\boldsymbol{C}$, wherein $\boldsymbol{C} := [\alpha_1(0:N-1), \alpha_2(0:N-1), \alpha_3(0:N-1), p(0:k-1)]$ and $\boldsymbol{I} := [x(0:N), x_1(0:N), x_2(0:N), x_3(0:N), y_1(0:N), y_2(0:N), y_3(0:N), \varphi(0:N), \varphi_1(0:N), \varphi_2(0:N), \varphi_3(0:N), \delta_r(0:N-1), \delta_{r1}(0:N-1), \delta_{r2}(0:N-1), \Delta d]$. The coarse path points can serve as initial guesses for the MIP problem and specify the boundary values for Big M method. Solving MIP can provide both the optimal sequence of motion modes decision and the optimal path simultaneously.

TABLE I: Definition of comparable algorithms.

Alg. ID	Description
HA	Path planning by hybrid A* method [9].
MA	The proposed modified A* method.

TABLE II: Parameter settings.

Parameters	Description	value
L_f/L_r	The CoG-to-front/rear axis length.	1.25m/1.66m
L_{fh}/L_{fh}	The front/rear overhang length.	0.5m/0.5m
W	The vehicle's width.	1.675m
N	The number of discrete path points.	20
$g_1/g_2/g_3$	The constants in (2).	1/1.7/2
$\beta_x/\beta_y/\beta_\varphi$	The constants in (16).	3m/3m/$\frac{\pi}{2}$
γ	The constant in (24).	$\frac{\pi}{4}$
$\lambda_1/\lambda_2/\lambda_3/\lambda_4$	The weighting of cost functions.	2/1/0.5/10

IV. Simulation results and Discussion

In this section, the performance of our proposed algorithm is validated through autonomous parking simulations in three representative cases: 1) parallel parking; 2) vertical parking; 3) tunnel parking. The comparable algorithms are defined in Table I.

The MIP problem is solved using the Gurobi optimizer, which can theoretically guarantees global optimality [20]. All generalized nonlinear constraints are approximated using piecewise linear functions, while the MIP problem formulated in this paper is primarily tackled using the branch and bound method. The simulations are implemented on a computer with an Intel(R) Core(TM) i5-11260H CPU @ 2.60GHz. The parameter settings are shown in Table II.

The simulation results of different methods in three scenarios are illustrated in Fig. 5, and the performance of each algorithm is summarized in Table III. The optimal motion modes decision sequences generated by the proposed method in three cases are shown in Fig. 6.

- *HA*: As shown in Fig. 5(a)-5(c), the HA method can successfully generate paths, but it exhibits frequent direction changes in narrow areas. Additionally, due to the inability of HA to guarantee optimality, the total path length is relatively long.
- *MA*: As shown in Fig. 5(d)-5(f), by incorporating the heading angle into the node state and redefining the node expansion rules, the MA method is capable of generating paths with motion mode decisions. However, the paths are not smooth and may involve unnecessary turns or mode switches. For example, in Case 2, the vehicle frequently moves forward and backward near $x = 9.5m$ under the diagonal motion mode; and in Case 3, the vehicle make unnecessary turns near $x = 10m$.
- *Ours*: As illustrated in Fig. 5(g)-5(i), the proposed algorithms in these three cases are capable of generating the

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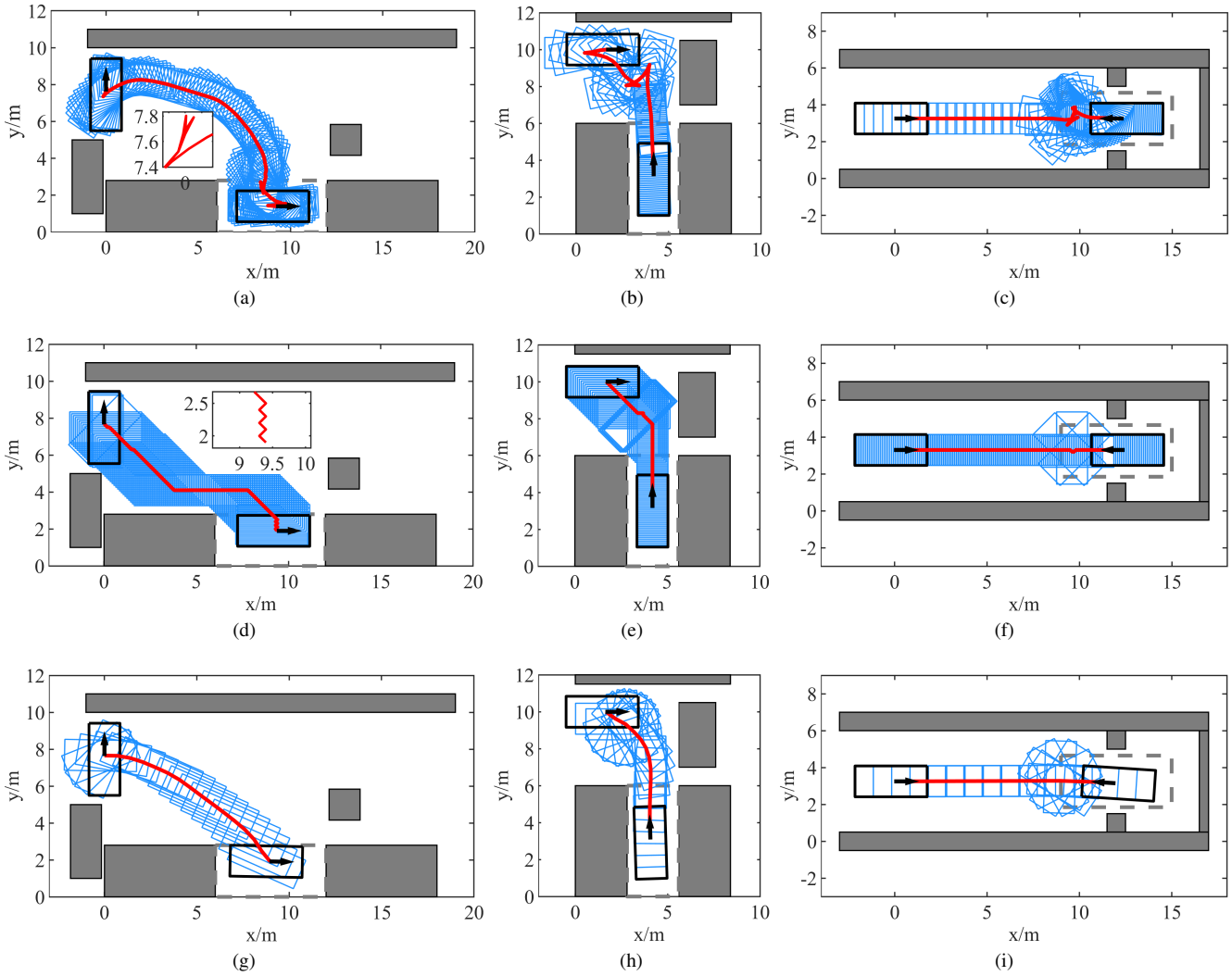


Fig. 5: Illustration of autonomous parking simulation results in narrow scenarios. (a)-(c) The path planning results using the HA method exhibit frequent changes in motion direction; (d)-(f) The path planning results using the MA method indicate the capability of mode switching but cannot guarantee optimality; (g)-(i) The path planning results using the proposed method provides the optimal paths and motion modes sequences.

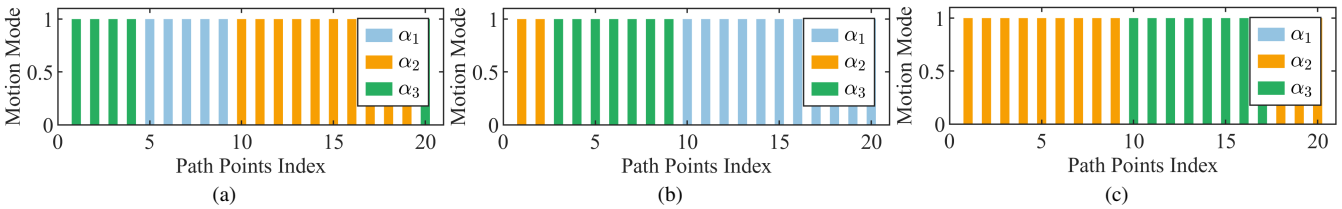


Fig. 6: The motion modes sequences in three cases of the proposed method. $\alpha_1 = 1$, $\alpha_2 = 1$, and $\alpha_3 = 1$ denote that the vehicle is moving under Ackermann steering mode, diagonal motion mode, and zero turn mode, respectively.

shortest and smoothest paths with appropriate motion modes sequences. Compared to the MA method, the modes sequences generated by the proposed algorithms exhibit slight differences, indicating that the optimizer has conducted more optimal exploration based on initial guesses. Furthermore, the generated paths fully adhere to the vehicle kinematics, a capability lacking in the MA algorithm.

The proposed algorithm effectively leverages the high flexibility and maneuverability advantages of 4WIS vehicles. This is attributed to the preliminary exploration of motion modes and paths during the coarse path planning stage,

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TABLE III: Performance comparison of different methods.

Method	Path length / m			Mode or direction switches times		
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
HA	17.9031	13.3284	15.9841	8	4	7
MA	12.0461	7.9752	12.4471	8	4	4
Ours	**10.9877**	**7.9238**	**11.9257**	**3**	**2**	**2**

along with the consideration of multiple motion modes and comprehensive costs in the formulated MIP problem. Note that the algorithm exhibits a relatively longer computational time as it provides not only the optimal path but also the optimal motion decision sequence. This work focuses on proposing a path planning framework for 4WIS vehicles rather than on methods for rapid solving of MIP problem, which can be explored in future research.

V. CONCLUSIONS

This work has proposed a novel optimization-based parking path planning approach integrating motion mode decision-making for 4WIS vehicles. In the coarse path search stage, an MA method has been proposed for preparation. In the optimal path planning stage, an optimization problem considering multiple kinematic characteristics of 4WIS vehicles has been formulated to generate optimal paths and motion mode sequences. The comparative simulations demonstrates that the proposed approach effectively leverages the flexibility and maneuverability of 4WIS vehicles, exhibiting the capability for efficient parking planning in narrow environments.

Future research will focus on velocity planning and smooth mode switch control. Additionally, accelerating the solution of MIP problems to satisfy real-time requirements will be considered.

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